	0	1	1	0
V2	0	0	0	1
V3	0	0	0	0
V4	0	0	0	0

As you can see, they are very different. One aspect that stands out immediately is that, in the adjacency matrix, most cells are filled with 0. This stems from the fact that the graph in figure 14.2 only has a few edges, of the many possible.

We know that, since edges are pairs of vertices, in a simple graph the maximum number of edges is $O(|V|2)$; what's the minimum number though?

It can be anything, a graph can even (hypothetically) have no edges at all; a connected graph, however, must have at least $|V|-1$ edges.

Table 14.1 Weakness and strengths of graphs representations

Operation	Adjacency List	Adjacency Matrix
Edge search/insert/delete	$O(\|E\|/\|V\|)$	$O(1)$
List of outgoing edges	$O(\|E\|/\|V\|)$	$O(\|V\|)$
List of ingoing edges	$O(\|E\|)$	$O(\|V\|)$
Space needed	$O(\|E\|+\|V\|)$	$O(\|V\|^2)$
Vertex insert	$O(1)$	$O(\|V\|)^{\S}$
Vertex delete	$O(\|E\|)^{\$}$	$O(\|V\|^2)$

($) We need to check all edges, to remove those whose destination has been deleted.